

Creativity in Combinatorial Configurations

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1 Intro

Both in olympiad questions and in research, creativity is needed for examples or counterexamples to statements. When an extremum is asked, we even further need to have an optimal construction and an argument why something is optimal.

Hereby, it turns out that in general, people are pretty bad in finding constructions. It took many years, before a mathematical hobbyist found an aperiodic monotile (known as the hat or an einstein). We also have for example the following probabilistic result on the Ramsey number, which we define for convenience of the reader.

Definition 1.1. *The Ramsey number*

$$R(k, m) = \min\{n \mid \text{every red/blue-colouring of } K_n \text{ has a red } K_k \text{ or blue } K_m\}.$$

Theorem 1.2. *For every $m > 3$, $R(m, m) > \sqrt{2}^m$.*

Proof. Let n satisfy $\binom{n}{m} 2^{1-\binom{m}{2}} < 1$. Then there is a colouring of the edges of K_n with two colours without monochromatic K_m . $\square$

On the other hand, there is no known construction certifying $R(m, m) > (1 + \epsilon)^m$ for all large m , for any $\epsilon > 0$.¹ The latter is a problem by Erdős for which he awards \$ 100.

So finding a construction of something that works in 100% of the cases, turns out to be very hard! Sometimes, people have the feeling that something should be true, even while they cannot fully resolve some simple cases.

Problem 1.3. *In $\mathbb{Z}^d$, one can consider tiles formed by $2k + 1$ consecutive points (parallel to one of the axes) with the central point removed. For every k , what is the least d for which a tiling of $\mathbb{Z}^d$ exists?*

Answer 1.4. *It turns out to be $d = \min\{k, 3\}$. For $k \leq 2$, this is easy. For $k \geq 3$, it is a case analysis that $d = 2$ does not work and a construction is needed that $d = 3$ is sufficient. The latter happened in multiple steps; a lemma reducing the problem to a more manageable 2D problem, and a construction which depends on $v_2(k)$. So in principle, infinitely many different constructions were formed to resolve the problem.*

Another simple posed question in graph theory, where one is looking for an example, did not result in any progress either.

Question 1.5 (Šoltes). *There is a graph, besides C_{11} , for which the total distance does not change by removing any of its vertices?*

When extending to generalisations (weighted graphs/ uniform hypergraphs/ digraphs), infinitely many constructions are known. Here one has found some concrete solutions among the plentitude of possibilities. Even if this is true for the graph case, one has to find a second needle among infinitely many, in an infinite hay stack.

Interestingly, many mathematical conjectures turn out to be false. Nowadays, computer generation and also machine learning helps to provide counterexamples.

¹First constructions lead to $R(m, m) > (m - 1)^2$.

2 Guidelines/ strategies

1. Try the problem for small values. Small values help you guess the minimum/maximum, and constructions for small values may generalize to general constructions.
2. Think simply, think symmetrically (except if the question suggest an assymetric aspect). It's rare that a problem has a complicated construction.
3. Make drawings, related tables or graphs (function or graph theory), and connect with different settings to get inspiration.
4. If the answer asks about existence; the answer may be both yes and no. So switch if you do not get there in one direction, and keep track of the properties that are derived during the search in one direction, as it may help for the other direction.
5. Derive properties and state them as claims. This can help with finding a construction, or as steps towards a proof of non-existence.

2.1 Methods

- induction: one of the most standard ways to derive a general construction
- picture; make drawings (in an abstract way) and try to observe
- probabilistic method; use random constructions (for part) and prove that with nonzero probability something exists
- greedy algorithms: work in a rough way/ straightforward way

One can also think about more classical techniques where the challenge is to construct it, e.g.

- a (semi-)invariant
- colouring of board
- finding a suitable tile

3 Useful constructions

In some cases, knowing some existing constructions with particular properties can help. The list below gives some examples (not all are useful for olympiads).

- analysis: particular sequences like the Farey sequence, functions with special properties like the Weierstrass function (continuous everywhere, but nowhere differentiable)
- algebra: identities, e.g. form of Pythagorean triples
- geometry: construction an auxiliary line or helpful additional point, knowing Euler line or some standard configurations
- graphs with arbitrary large girth and chromatic number (e.g. Mycielskian graphs)
- projective planes: sets of $q^2 + q + 1$ points (each belonging to $q + 1$ lines) and lines (each contain $q + 1$ points). These exist for every prime power $q = p^k$, but it is unknown if no other example exists (some small values are excluded).
- biplanes; a set $\{1, 2, \dots, n\}$ of elements (points), together with blocks (lines), such that every two points belong to exactly two common blocks, and every two blocks share exactly two points (known to exist for some small $k \leq 13$, but unknown if there are infinitely many)
- Hamming codes: maximum subset of $\{0, 1\}^n$ for $n = 2^r - 1$ which is error-correcting with message length $k = 2^r - r - 1$. The subset contains 2^k elements and every two different strings are different in at least 3 entries
- number theory: (density) of primes, d -ary representation

Nevertheless, in olympiads the constructions can be as simple as consecutive integers in a certain range, or points on a regular n -gon. It is thus hard to give waterproof general guidelines, and hence practicing and creating an own feeling + thinking creatively at the moment is all one can do.

4 Example olympiad questions

The fact that constructions are important at IMO, can be derived from the following (incomplete) list:
IMO 2010 Q5, IMO 2011 Q2, IMO 2012 Q6, IMO 2013 Q1, IMO 2014 Q2, IMO 2015 Q1, IMO 2016 Q1, or more recently IMO 2024 Q5.

✓ IMO '77, VWO '10 In a finite sequence of real numbers the sum of any seven successive terms is negative and the sum of any eleven successive terms is positive. Determine the maximum number of terms in the sequence.

✓ IMC '13 There are $2n$ students in a school ($n \geq 2$ is an integer). Each week n students go on a trip. After several trips the following condition was fulfilled: every two students were together on at least one trip. What is the minimum number of trips needed for this to happen?

✓ JEMC '23 Q2 Let $n \geq 5$ be an integer. There are n points in the plane, no three of them collinear. Each day, Tom erases one of the points, until there are three points left. On the i -th day, for $1 \leq i \leq n-3$, before erasing that day's point, Tom writes down the positive integer $v(i)$ such that the convex hull of the points at that moment has $v(i)$ vertices. Finally, he writes down $v(n-2) = 3$. Find the greatest possible value that the expression

$$|v(1) - v(2)| + |v(2) - v(3)| + \dots + |v(n-3) - v(n-2)|$$

can obtain among all possible initial configurations of n points and all possible Tom's moves.

Remark. A convex hull of a finite set of points in the plane is the smallest convex polygon containing all the points of the set (inside it or on the boundary).

✓ USAMO '14 Prove that there exists an infinite set of points

$$\dots, P_{-3}, P_{-2}, P_{-1}, P_0, P_1, P_2, P_3, \dots$$

in the plane with the following property: For any three distinct integers a, b , and c , points P_a, P_b , and P_c are collinear if and only if $a + b + c = 2014$.

✓ EGMO '23 Q3 Let k be a positive integer. Lexi has a dictionary D consisting of some k -letter strings containing only the letters A and B. Lexi would like to write either the letter A or the letter B in each cell of a $k \times k$ grid so that each column contains a string from D when read from top-to-bottom and each row contains a string from D when read from left-to-right. What is the smallest integer m such that if D contains at least m different strings, then Lexi can fill her grid in this manner, no matter what strings are in D ?

✓ IMC '13 Does there exist an infinite set M consisting of positive integers such that for any $a, b \in M$, with $a < b$, the sum $a + b$ is square-free?

5 Solution Sketches

IMO '77, VWO '10 In a finite sequence of real numbers the sum of any seven successive terms is negative and the sum of any eleven successive terms is positive. Determine the maximum number of terms in the sequence.

Answer 5.1. Here one can play with the negative and positive sums, and replace $(a, b) = (11, 7)$ first by smaller values (relative prime) to see what is going on. If the sequence has length at least 17, every four consecutive terms can be extended by putting 7 next to it (and needs to have positive sum). Thus one repeats with $(7, 4), (4, 3), (3, 1)$ and concludes this is impossible.

The more elegant way that a sequence $(x_i)_i$ of length 17 cannot exist, is considering

$$\sum_{i=1..11} \sum_{j=0..6} x_{i+j} = \sum_{j=1..7} \sum_{i=0..10} x_{i+j}$$

which has to be both positive and negative, impossible.

In the other direction, aiming to find an example of length 16, we know the central 4 need to have a negative sum (and puzzling a bit more, being $-++-$ if not mistaken). If all positive - and all negative values should have fixed value, we want the other four tuples to have only one negative term, and indeed.

Since $\frac{2}{7} > \frac{3}{11}$, one can do

ppnppppnppnpppp

where $p > 0$ and $n < 0$ are negative values satisfying $5p + 2n < 0$ and $8p + 3n > 0$.

IMC '13 There are $2n$ students in a school ($n \geq 2$ is an integer). Each week n students go on a trip. After several trips the following condition was fulfilled: every two students were together on at least one trip. What is the minimum number of trips needed for this to happen?

Answer 5.2. Since every student needs to attend at least 3 trips, by double counting at least 6 trips are needed.

Next, one ensures that 6 is tight by a construction. When $2 \mid n$, this is done by 4 groups A, B, C, D of size $\frac{n}{2}$ and pairing them in all possible ways (one can think as the edges of a K_4 , with the vertices those groups).

If $n = 3$, call the 6 people E, F, G, H, I, J . Then the members of trips may be the following: $(E, F, G), (E, F, H), (G, H, I), (G, H, J), (E, I, J), (F, I, J)$.

Write $n = 2x + 3$ for odd $n > 3$ and combine the constructions of $n = 2x$ and $n = 3$;

$(A, B, E, F, G), (C, D, E, F, H), (A, C, G, H, I)$

$(B, D, G, H, J), (A, D, E, I, J), (B, C, F, I, J)$.

JEMC '23 Q2 Let $n \geq 5$ be an integer. There are n points in the plane, no three of them collinear. Each day, Tom erases one of the points, until there are three points left. On the i -th day, for $1 \leq i \leq n - 3$, before erasing that day's point, Tom writes down the positive integer $v(i)$ such that the convex hull of the points at that moment has $v(i)$ vertices. Finally, he writes down $v(n - 2) = 3$. Find the greatest possible value that the expression

$$|v(1) - v(2)| + |v(2) - v(3)| + \dots + |v(n - 3) - v(n - 2)|$$

can obtain among all possible initial configurations of n points and all possible Tom's moves.

Remark. A convex hull of a finite set of points in the plane is the smallest convex polygon containing all the points of the set (inside it or on the boundary).

Answer 5.3. Every vertex can join and leave the convex hull, and contribute at most 2. The first ≥ 3 and last 3 vertices cannot contribute for entering resp. leaving. Hence $2n - 6$ is already an upper bound, and if we remove a vertex not belonging to the current convex hull, we already have a bound of $2(n - 1) - 6 = 2n - 8$. If a vertex of the current convex hull is removed everytime, a vertex needs to enter and to be removed at the same time, implying a loss of 2, or there were no vertices not on the convex hull and the bound is $n - 3 \leq 2n - 8$. In all cases, we have an upper bound of $2n - 8$.

Equality is obtained by having a convex polygon of order $n - 1$, inscribed in a triangle, which shares two vertices with the polygon. (first term $n - 4$, remaining $n - 4$ terms are 1)

USAMO '14 Prove that there exists an infinite set of points

$$\dots, P_{-3}, P_{-2}, P_{-1}, P_0, P_1, P_2, P_3, \dots$$

in the plane with the following property: For any three distinct integers a, b , and c , points P_a, P_b , and P_c are collinear if and only if $a + b + c = 2014$.

Answer 5.4. Using the Cartesian coordinate system, and letting $P_x = (x, f(x))$ have x -coordinate x , we want that the line through any two points P_a and P_b , if $2024 - (a + b) \notin \{a, b\}$, goes through the third point P_c , $c = 2024 - (a + b)$. If f is a polynomial, we need at least a third degree polynomial. In that case, the line $y = xc + d$ intersects $y = f(x)$ in three points, and the roots of $f(x) - cx - d = 0$ need to sum to zero. By Vieta juming, $f(x)$ is a multiple of $x^3 - 2014x^2 + rx + s$.

EGMO '23 Q3 Let k be a positive integer. Lexi has a dictionary D consisting of some k -letter strings containing only the letters A and B . Lexi would like to write either the letter A or the letter B in each cell of a $k \times k$ grid so that each column contains a string from D when read from top-to-bottom and each row contains a string from D when read from left-to-right. What is the smallest integer m such that if D contains at least m different strings, then Lexi can fill her grid in this manner, no matter what strings are in D ?

Answer 5.5. The minimum is 2^{k-1} .

The construction of all k -letter strings ending in B different from the one with only B s, is clearly an example for which the grid cannot be filled in the desired manner.

If the pure A string, or pure B string are in D , the matrix with that one letter satisfies the condition. In the other case, with $m = 2^{k-1}$ strings, there are two complementary strings (every index has the opposite letter). Now we can fill the grid with only those two strings. Working with 0 and 1s instead, one could write the string as $R_1 R_2 \dots R_k$ and $C_1 C_2 \dots C_k$, and put $R_i + C_j \pmod 2$ in cell (i, j) .

IMC '13 Does there exist an infinite set M consisting of positive integers such that for any $a, b \in M$, with $a < b$, the sum $a + b$ is square-free?

Answer 5.6. We construct M recursively, by adding the positive integers greedily as long as they satisfy the condition. Then $M = \{1, 2, 4, 9, 13, 29, \dots\}$. All we need to check is that M is indeed an infinite set. So assume it is finite, $\{m_1 = 1, \dots, m_k\}$. Let $y = ((\sum m_k)!)^2$. We now consider elements of the form $1 + xy$. m_i cannot be a multiple of p^2 for any prime $p \leq \sum m_k$ by construction and assumption. For $1 \leq x \leq N$ for a large value N , the elements $1 + xy + m_i$ can only possibly be a multiple of p^2 where $2k < \sum m_k < p < \sqrt{(N+1)y}$. The number of values x for which this happens for at least one m_i is bounded by

$$\sum_{2k < p < \sqrt{(N+1)y}}^k \left(1 + \frac{N}{p^2}\right).$$

The latter is smaller than $\frac{N}{2} + k\sqrt{(N+1)y} < N$ for large N and thus there is a valid choice of the form $1 + xy$, from which we conclude.